

# User Manual

## ICAM-300 Series

### Integrated Industrial AI Camera

**ADVANTECH**

*Enabling an Intelligent Planet*

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## Product Warranty (2 years)

Advantech warrants the original purchaser that each of its products will be free from defects in materials and workmanship for two years from the date of purchase.

This warranty does not apply to any products that have been repaired or altered by persons other than repair personnel authorized by Advantech, or products that have been subject to misuse, abuse, accident, or improper installation. Advantech assumes no liability under the terms of this warranty as a consequence of such events.

Because of Advantech's high quality-control standards and rigorous testing, most customers never need to use our repair service. If an Advantech product is defective, it will be repaired or replaced free of charge during the warranty period. For out-of-warranty repairs, customers will be billed according to the cost of replacement materials, service time, and freight. Please consult your dealer for more details.

If you believe your product to be defective, follow the steps outlined below.

1. Collect all the information about the problem encountered. (For example, CPU speed, Advantech products used, other hardware and software used, etc.) Note anything abnormal and list any onscreen messages displayed when the problem occurs.
2. Call your dealer and describe the problem. Please have your manual, product, and any helpful information readily available.
3. If your product is diagnosed as defective, obtain a return merchandise authorization (RMA) number from your dealer. This allows us to process your return more quickly.
4. Carefully pack the defective product, a completed Repair and Replacement Order Card, and a proof of purchase date (such as a photocopy of your sales receipt) into a shippable container. Products returned without a proof of purchase date are not eligible for warranty service.
5. Write the RMA number clearly on the outside of the package and ship the package prepaid to your dealer.

# Declaration of Conformity

## CE

This product has passed the CE test for environmental specifications when shielded cables are used for external wiring. We recommend the use of shielded cables. This type of cable is available from Advantech. Please contact your local supplier for ordering information.

Test conditions for passing also include the equipment being operated within an industrial enclosure. In order to protect the product from damage caused by electrostatic discharge (ESD) and EMI leakage, we strongly recommend the use of CE-compliant industrial enclosure products.

## FCC Class A

Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at their own expense.

## FM

This equipment has passed the FM certification. According to the National Fire Protection Association, work sites are classified into different classes, divisions and groups, based on hazard considerations. This equipment is compliant with the specifications of Class I, Division 2, Groups A, B, C and D indoor hazards.

# Technical Support and Assistance

1. Visit the Advantech website at [www.advantech.com/support](http://www.advantech.com/support) to obtain the latest product information.
2. Contact your distributor, sales representative, or Advantech's customer service center for technical support if you need additional assistance. Please have the following information ready before calling:
  - Product name and serial number
  - Description of your peripheral attachments
  - Description of your software (operating system, version, application software, etc.)
  - A complete description of the problem
  - The exact wording of any error messages

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## Warnings, Cautions and Notes

**Warning!** *Warnings indicate conditions, which if not observed, can cause personal injury!*



**Caution!** *Cautions are included to help prevent hardware damage and data losses. For example,*



*“Batteries are at risk of exploding if incorrectly installed. Do not attempt to recharge, force open, or heat the battery. Replace the battery only with the same or equivalent type as recommended by the manufacturer. Discard used batteries according to the manufacturer’s instructions.”*

**Note!** *Notes provide optional additional information.*



## Document Feedback

To assist us with improving this manual, we welcome all comments and constructive criticism. Please send all such feedback in writing to [support@advantech.com](mailto:support@advantech.com).

## Safety Precaution - Static Electricity

Follow these simple precautions to protect yourself from harm and the products from damage.

- To avoid electrical shock, always disconnect the power from the PC chassis before manual handling. Do not touch any components on the CPU card or other cards while the PC is powered on.
- Disconnect the power before making any configuration changes. A sudden rush of power after connecting a jumper or installing a card may damage sensitive electronic components.

## Safety Instructions

1. Read these safety instructions carefully.
2. Retain this user manual for future reference.
3. Disconnect the equipment from all power outlets before cleaning. Use only a damp cloth for cleaning. Do not use liquid or spray detergents.
4. For pluggable equipment, the power outlet socket must be located near the equipment and easily accessible.
5. Protect the equipment from humidity.
6. Place the equipment on a reliable surface during installation. Dropping or letting the equipment fall may cause damage.
7. The openings on the enclosure are for air convection. Protect the equipment from overheating. Do not cover the openings.
8. Ensure that the voltage of the power source is correct before connecting the equipment to a power outlet.
9. Position the power cord away from high-traffic areas. Do not place anything over the power cord.
10. All cautions and warnings on the equipment should be noted.
11. If the equipment is not used for a long time, disconnect it from the power source to avoid damage from transient overvoltage.
12. Never pour liquid into an opening. This may cause fire or electrical shock.
13. Never open the equipment. For safety reasons, the equipment should be opened only by qualified service personnel.
14. If any of the following occurs, have the equipment checked by service personnel:
  - The power cord or plug is damaged.
  - Liquid has penetrated the equipment.
  - The equipment has been exposed to moisture.
  - The equipment is malfunctioning, or does not operate according to the user manual.
  - The equipment has been dropped and damaged.
  - The equipment shows obvious signs of breakage.
15. Do not leave the equipment in an environment with a storage temperature of below -20 °C (-4 °F) or above 60 °C (140 °F) as this may damage the components. The equipment should be kept in a controlled environment.
16. CAUTION: Batteries are at risk of exploding if incorrectly replaced. Replace only with the same or equivalent type as recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.
17. In accordance with IEC 704-1:1982 specifications, the sound pressure level at the operator's position does not exceed 70 dB (A).

DISCLAIMER: These instructions are provided according to IEC 704-1 standards. Advantech disclaims all responsibility for the accuracy of any statements contained herein.

## Wichtige Sicherheitshinweise

1. Bitte lesen Sie sich diese Hinweise sorgfältig durch.
2. Heben Sie diese Anleitung für den späteren Gebrauch auf.
3. Vor jedem Reinigen ist das Gerät vom Stromnetz zu trennen. Verwenden Sie keine Flüssig- oder Aerosolreiniger. Am besten dient ein angefeuchtetes Tuch zur Reinigung.
4. Die Netzanschlusssteckdose soll nahe dem Gerät angebracht und leicht zugänglich sein.
5. Das Gerät ist vor Feuchtigkeit zu schützen.
6. Bei der Aufstellung des Gerätes ist auf sicheren Stand zu achten. Ein Kippen oder Fallen könnte Verletzungen hervorrufen.
7. Die Belüftungsöffnungen dienen zur Luftzirkulation, die das Gerät vor Überhitzung schützt. Sorgen Sie dafür, dass diese Öffnungen nicht abgedeckt werden.
8. Beachten Sie beim Anschluss an das Stromnetz die Anschlusswerte.
9. Verlegen Sie die Netzanschlussleitung so, dass niemand darüber fallen kann. Es sollte auch nichts auf der Leitung abgestellt werden.
10. Alle Hinweise und Warnungen, die sich an den Geräten befinden, sind zu beachten.
11. Wird das Gerät über einen längeren Zeitraum nicht benutzt, sollten Sie es vom Stromnetz trennen. Somit wird im Falle einer Überspannung eine Beschädigung vermieden.
12. Durch die Lüftungsöffnungen dürfen niemals Gegenstände oder Flüssigkeiten in das Gerät gelangen. Dies könnte einen Brand bzw. elektrischen Schlag auslösen.
13. Öffnen Sie niemals das Gerät. Das Gerät darf aus Gründen der elektrischen Sicherheit nur von autorisiertem Servicepersonal geöffnet werden.
14. Wenn folgende Situationen auftreten, ist das Gerät vom Stromnetz zu trennen und von einer qualifizierten Servicestelle zu überprüfen:
  - Netzkabel oder Netzstecker sind beschädigt.
  - Flüssigkeit ist in das Gerät eingedrungen.
  - Das Gerät war Feuchtigkeit ausgesetzt.
  - Wenn das Gerät nicht der Bedienungsanleitung entsprechend funktioniert oder Sie mit Hilfe dieser Anleitung keine Verbesserung erzielen.
  - Das Gerät ist gefallen und/oder das Gehäuse ist beschädigt.
  - Wenn das Gerät deutliche Anzeichen eines Defektes aufweist.
15. **VORSICHT:** Explosionsgefahr bei unsachgemäßen Austausch der Batterie. Ersatz nur durch denselben oder einem vom Hersteller empfohlenen ähnlichen Typ. Entsorgung gebrauchter Batterien nach Angaben des Herstellers.
16. **ACHTUNG:** Es besteht die Explosionsgefahr, falls die Batterie auf nicht fachmännische Weise gewechselt wird. Verfangen Sie die Batterie nur gleicher oder entsprechender Type, wie vom Hersteller empfohlen. Entsorgen Sie Batterien nach Anweisung des Herstellers.
17. Der arbeitsplatzbezogene Schalldruckpegel nach DIN 45 635 Teil 1000 beträgt 70dB(A) oder weniger.

Haftungsausschluss: Die Bedienungsanleitungen wurden entsprechend der IEC-704-1 erstellt. Advantech lehnt jegliche Verantwortung für die Richtigkeit der in diesem Zusammenhang getätigten Aussagen ab.

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# Chapter 1

## Product Overview

## 1.1 Introduction

The Advantech ICAM-300 Series is a lightweight, all-in-one Industrial AI camera designed for manufacturing, robotics, warehouse automation, and workplace safety. Equipped with a 5MP global shutter sensor, fast autofocus, and integrated LED illumination, it enables real-time object detection, recognition, safety monitoring, risk avoidance, pick & place, and guidance On-Device AI applications.

Powered by the Qualcomm® QCS6490 platform, the ICAM-300 delivers robust AI performance with low power consumption. Its highly integrated hardware and software framework, including an QIM SDK for multimedia and camera control, GStreamer for real-time video and AI pipelines, Qualcomm AI Hub for access to optimized AI models, and Advantech's CamNavi SDK, which reduces the learning curve, accelerates time to value.

The ICAM-300 combines a compact form factor with a rugged IP66-rated enclosure, featuring exceptional anti shock, vibration and fog, equipped with large corner LED indicators and optional Wi-Fi dongle and more, offering maximum reliability for next-generation Vision AI applications in manufacturing sector.

## 1.2 Solution Features

- High sensitivity, global shutter 5MP image sensor
- Fast Autofocus with Voice Coil Motor (VCM) lens and advanced LED Illumination
- High Performance, low power Qualcomm® QCS6490 AI Processor with longevity support
- Compatible with Qualcomm® Intelligent Multimedia SDK (QIM SDK), AI Hub, and GStreamer.
- Lightweight and IP66 compact housing for mobility and On-Device AI in harsh environment

## 1.3 Specifications

- **Sensor:** Sensor: Omnivision OG05B20 with full size resolution 2592(H) x 1944(V), 5MP @30FPS, 1/2.53", pixel size (2.2 x 2.2 um), Global Shutter, Color

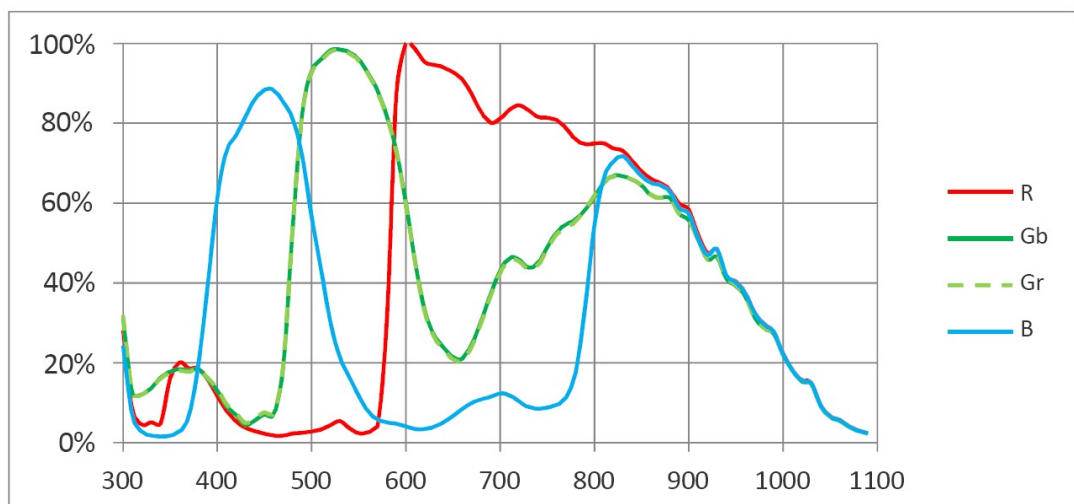
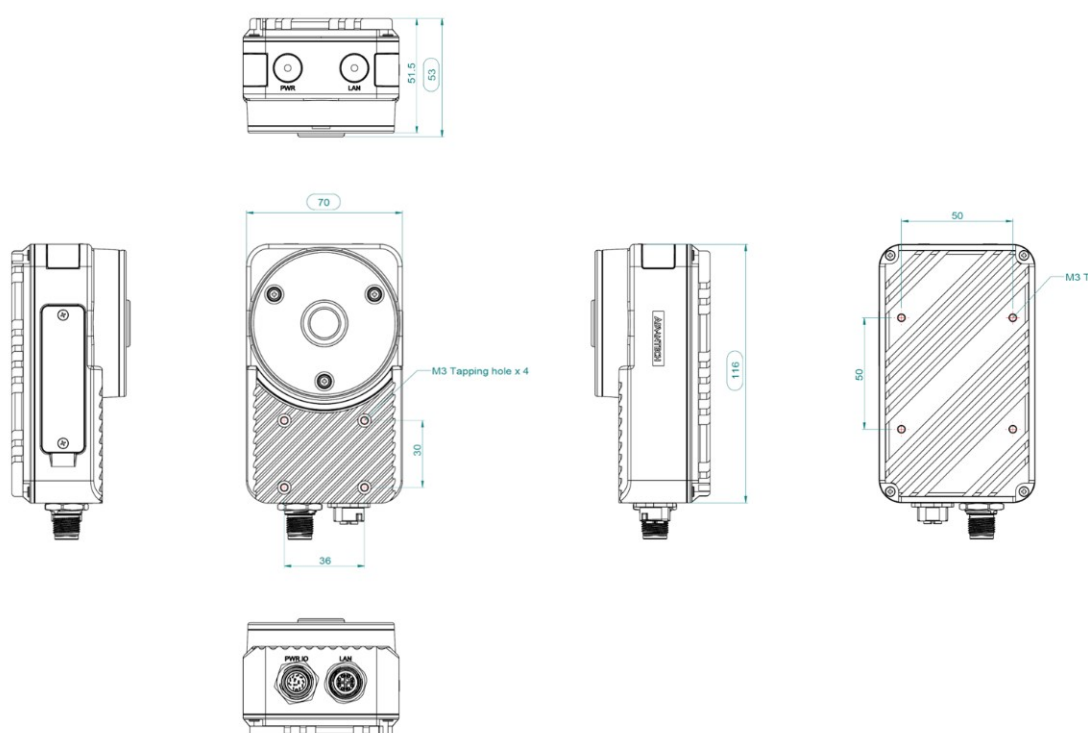


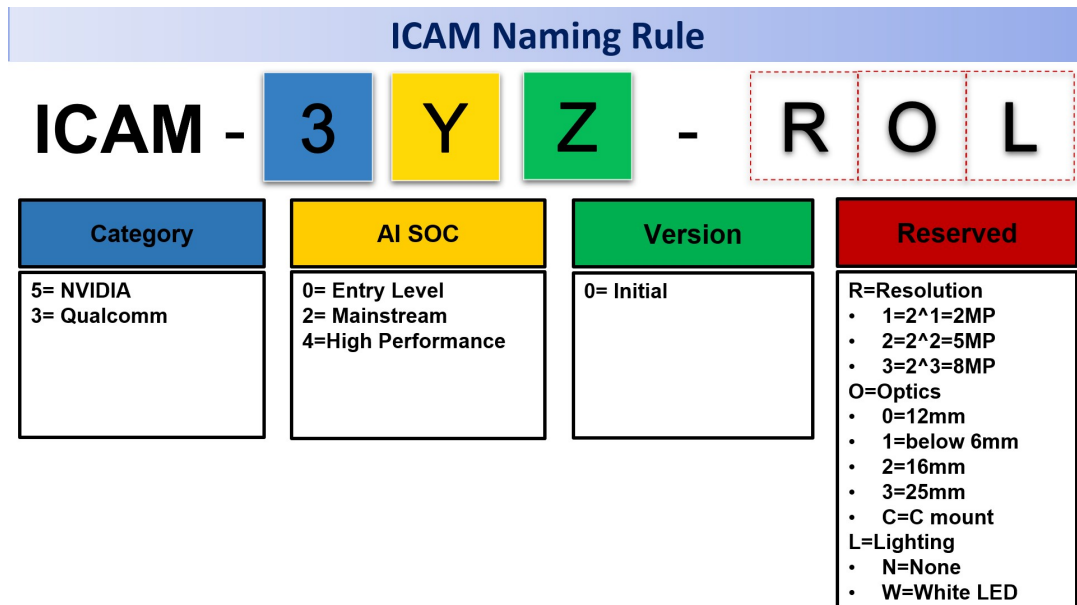
Figure 1.1 Spectrum response (nm)

- **Processor System:** Qualcomm QCS6490
  - AI Performance: Qualcomm 6th Gen AI Engine, up to 12 TOPS
  - CPU: 8-core Kryo 670 (ARM® v8.2, 64-bit), up to 2.1 GHz
  - GPU: Adreno 643L, up to 812 MHz; supports OpenGL® ES 3.2, Vulkan®, OpenCL™ 2.0, DirectX® 12
  - DSP: Qualcomm® Hexagon™ Processor
- **Memory, Storage:** 8 GB LPDDR5/128GB UFS
- **Optical:**
  - 6 mm variable focal length:
    - FOV 211.8 x 154.1 mm @200 mm working distance
    - FOV 1030.6 x 745.7 mm @1,000 mm working distance
  - LED illumination (S-mount model only) 8 x PWM white LEDs, programmable
- **Synchronization:** Software Trigger/Free-run
- **HW ISP:** Qualcomm Spectra™ ISP 570L
- **I/O:**
  - 1 x USB 3.0 Type A connector
  - 1 x Micro USB for OTG (Only works in recovery mode)
  - 2 x Digital Input
  - 1 x Digital Output
  - 1 x Digital Output / Strobe Output
- **LAN:** 1x 10/100/1000 Base-T
- **Power Requirements:** 19~24V<sub>DC</sub> Max 17W, typical 12W.
- **Dimension:** S-mount: Dimensions 70mm (W) x 116mm (H) x 53mm (D)



- **Software package:**
  - OS: Yocto based BSP (kernel 6.6 or above).
  - SDK/Utility: Neural Processing SDK (NPSDK), QIM, CAMNavi SDK, Web based camera utility

■ Naming Rule:



## 1.4 Environment Specifications

- Operating temperature: 0 ~ 45°C
- Vibration during operation: 5 Grms
- EMC: CE, FCC

# Chapter 2

## Optical Functions

## 2.1 Overview of Optical Functions

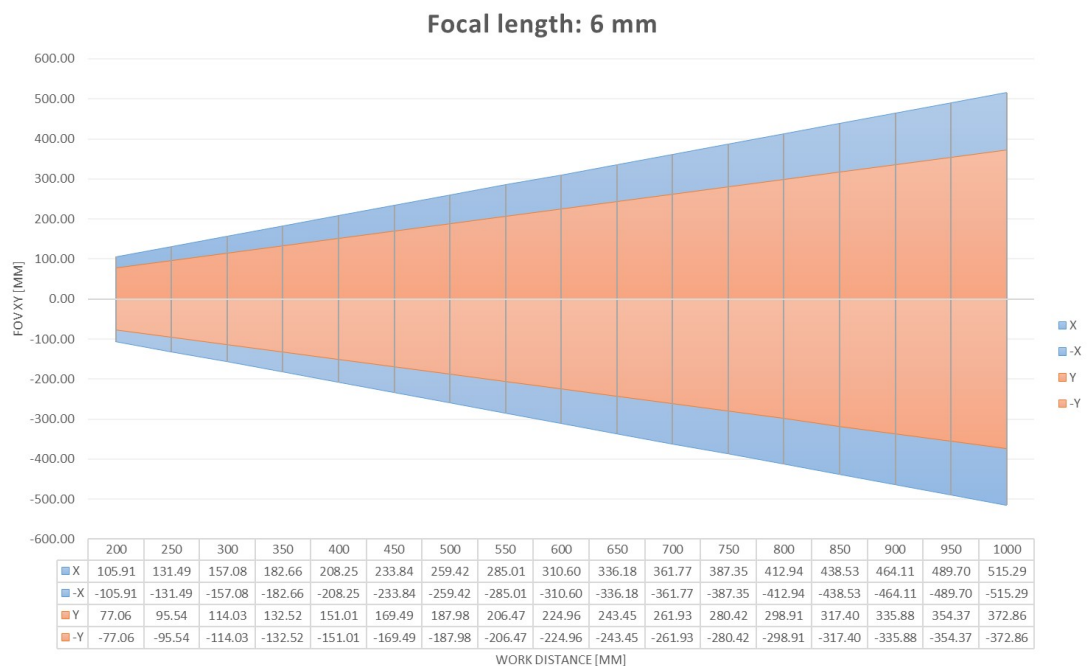
The Advantech ICAM-300 series is equipped with a highly integrated optical module, which includes built-in programmable variable focus lenses. The HTML 5 web based utility can be used to setup the optical parameters for application to lower the installation effort. This section shows optical functions of camera. This section shows the specification of optical function that included the field of view of lens, lens motor movement behavior, spectral distribution of LEDs and LEDs uniformity.

## 2.2 Field of View

**Table 2.1: S-mount Model**

| Focal length [mm] | FOV@Near<br>HxV [deg] | FOV@Near<br>HxV [mm] | FOV@Far<br>HxV [deg] | FOV@Far<br>HxV [mm] |
|-------------------|-----------------------|----------------------|----------------------|---------------------|
| 6                 | 54.2 x 40.6           | 211.8 x 154.1        | 54.2 x 40.6          | 1030.6 x 745.7      |
|                   | WD@200 mm             | WD@200 mm            | WD@1000 mm           | WD@1000 mm          |

WD: Working Distances; H x V: Horizontal x Vertical

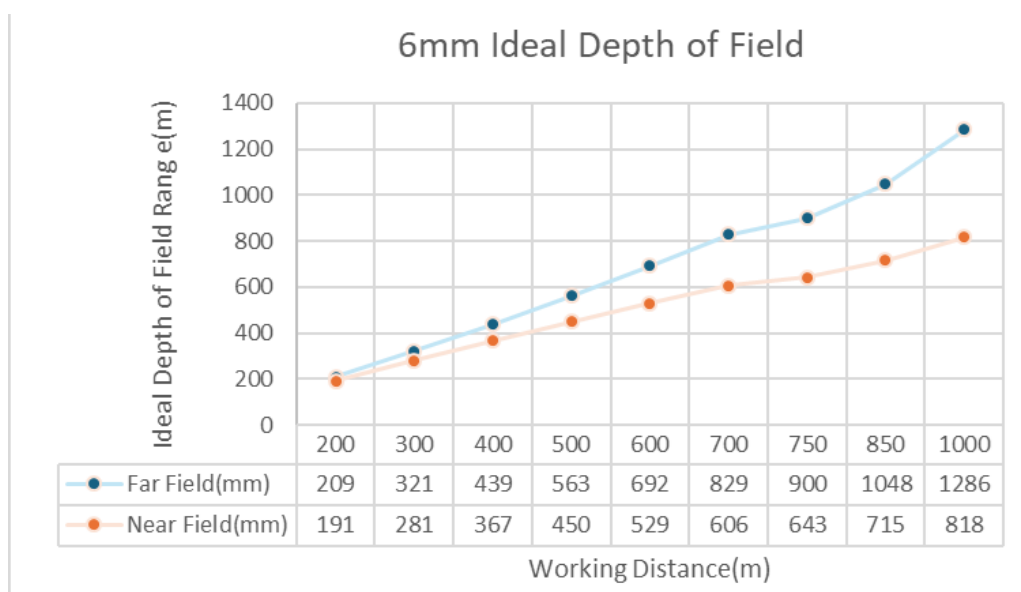


## 2.3 Ideal Depth of Field

**Table 2.2: S-mount Model**

| Focal length<br>[mm] | DOF@<br>Far Field [mm] | DOF@<br>Near Field [mm] | DOF@<br>Far Field [mm] | DOF@<br>Near Field [mm] |
|----------------------|------------------------|-------------------------|------------------------|-------------------------|
| 6                    | 209                    | 191                     | 1286                   | 818                     |
|                      | WD@200 mm              | WD@200 mm               | WD@1000 mm             | WD@1000 mm              |

WD: Working Distances



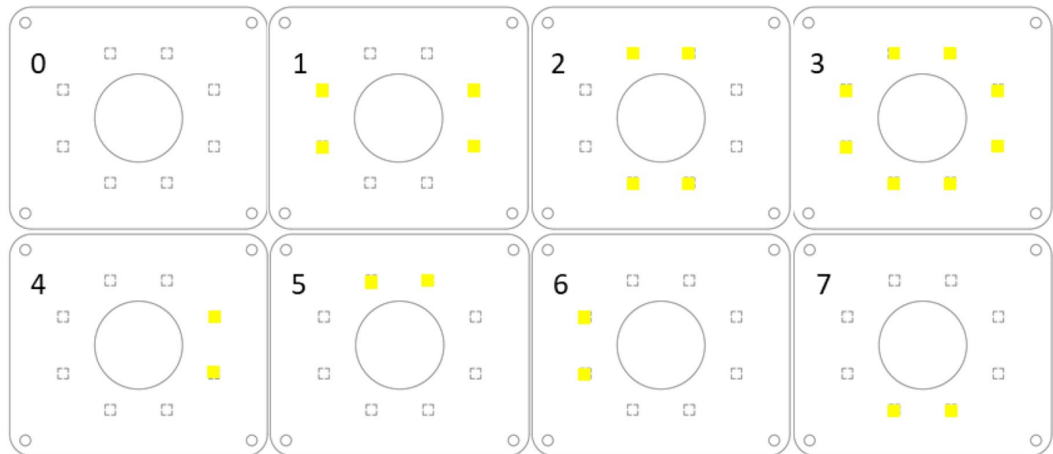
## 2.4 Lens Movement

The ICAM-300 series is a closed-loop VCM. It delivers high-performance auto focus, moving from far to near in approximately 2 seconds with a very short response time.

**Table 2.3: Specifications**

|                            |                 |
|----------------------------|-----------------|
| AF VCM type                | close-loop VCM. |
| Working distance range     | 200 ~ 1000 mm   |
| Lens position range        | 0 ~ 399         |
| AF speed (Office lighting) | < 1.5 s         |

## 2.5 LED Control Model



# Chapter 3

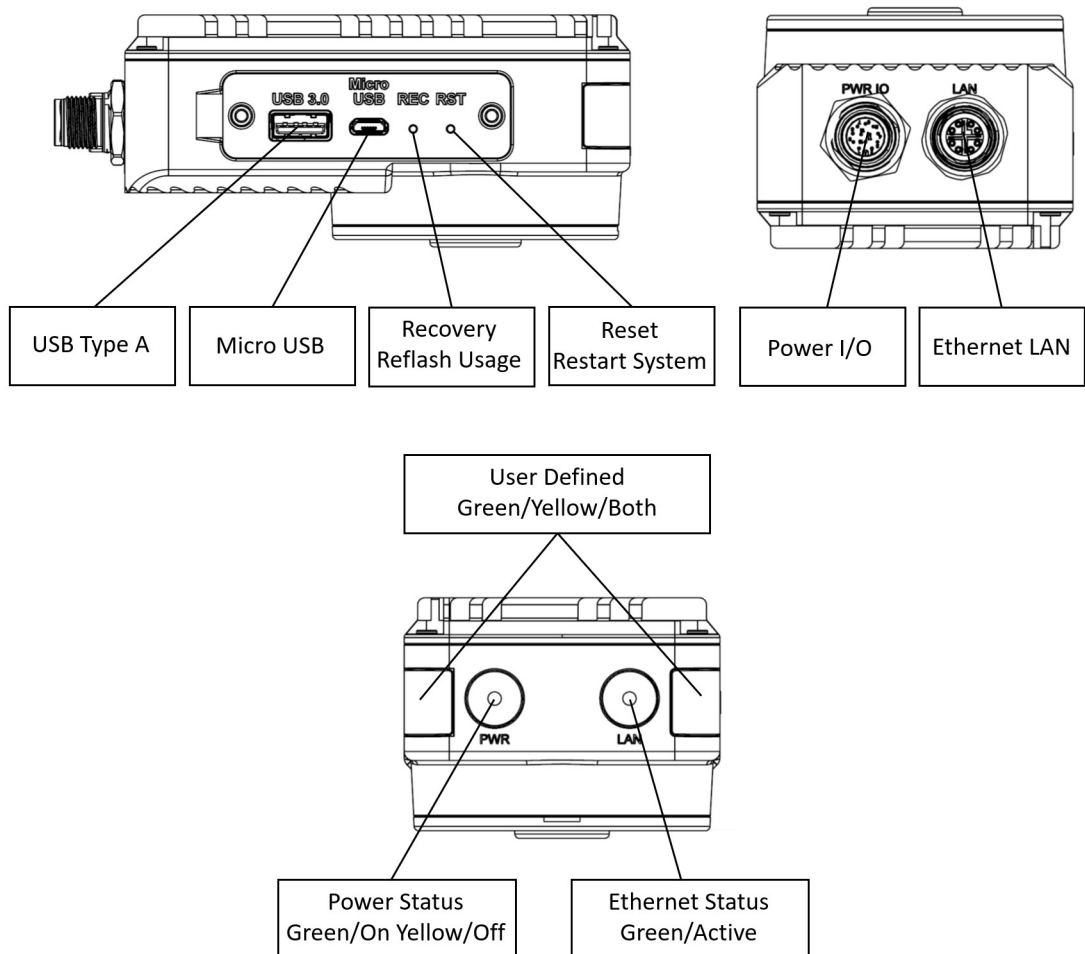
## Hardware Installation

## 3.1 Hardware Installation

The electrical installation of the ICAM-300 must be carried out by a qualified electrician. Disconnect all electrical components from the power supply when installing the ICAM-300. The following sections show the external connectors and pin assignments for applications.

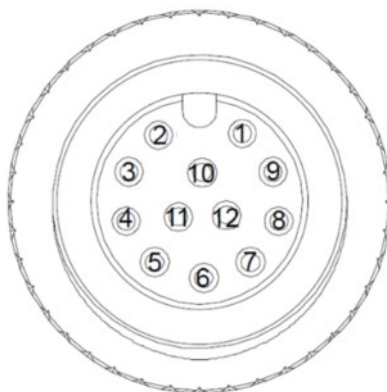
## 3.2 External Connector

### 3.2.1 External I/O Connector Indicator



### 3.2.2 Power I/O

Power, Digital I/O, and RS232 Connector

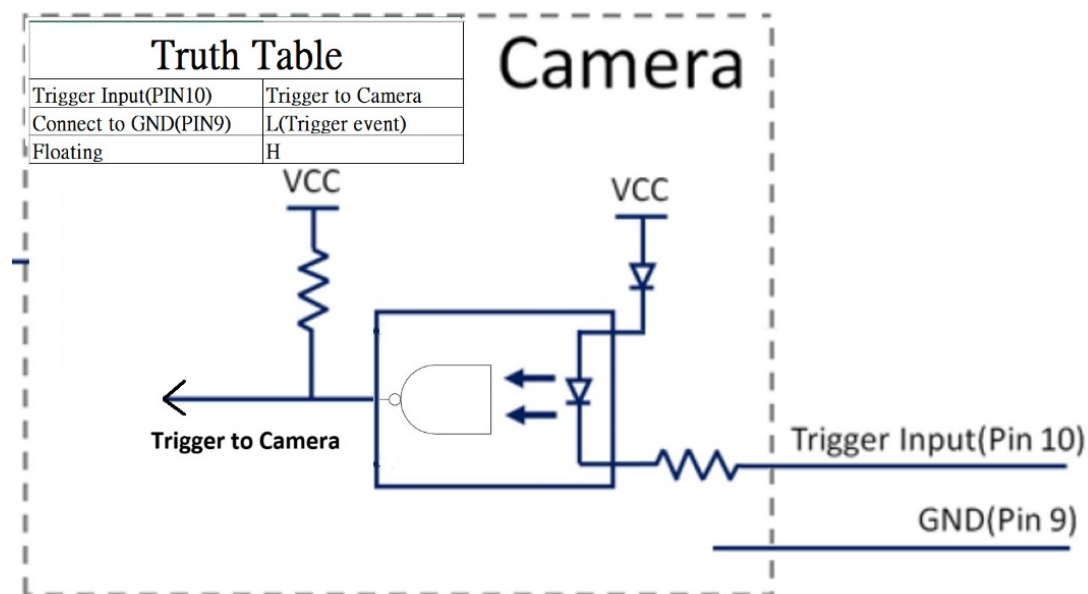


**Table 3.1: Power I/O**

| Pin | Description                   |
|-----|-------------------------------|
| 1   | Digital Input 1               |
| 2   | Reserve                       |
| 3   | Reserve                       |
| 4   | Reset System BTN              |
| 5   | Digital Input 0               |
| 6   | Digital Input Common          |
| 7   | System power in Vin(+) 24V    |
| 8   | System power in GND Vin(-)    |
| 9   | Digital GND                   |
| 10  | Reserve                       |
| 11  | Digital Output 0              |
| 12  | Digital Output 1 / Strobe Out |

ICAM-300 series supports versatile digital I/O. Please read the following instructions to wire the DI/O.

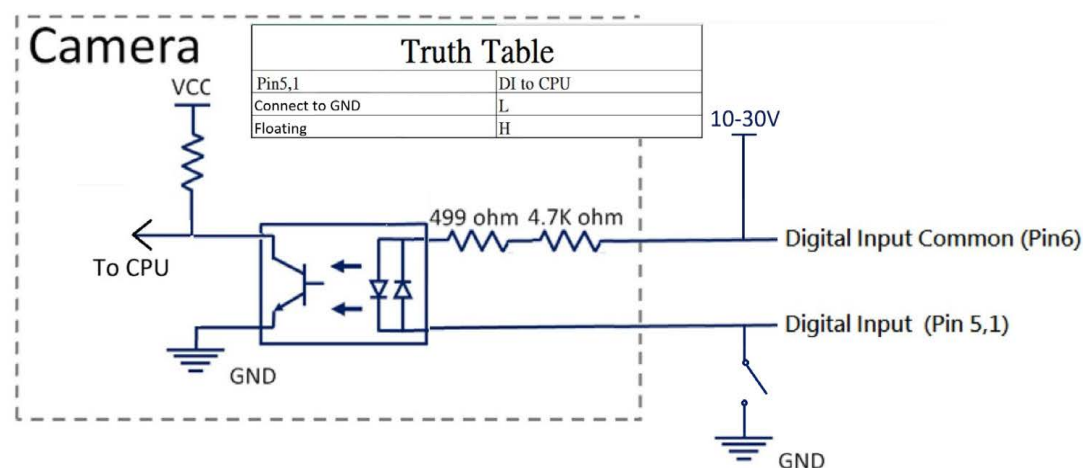
| Pin No. | Duration  | S/W Interrupt |
|---------|-----------|---------------|
| 10      | 0-8192 us | No            |



| Pin No. | Input Voltage                 | Input Current | Frequency  | Response Time | S/W Interrupt |
|---------|-------------------------------|---------------|------------|---------------|---------------|
| 5, 1    | High: 10-30Vdc<br>Low: 0-3Vdc | Max: 10mA     | Max. 60 Hz | Max: 18.8us   | Yes           |

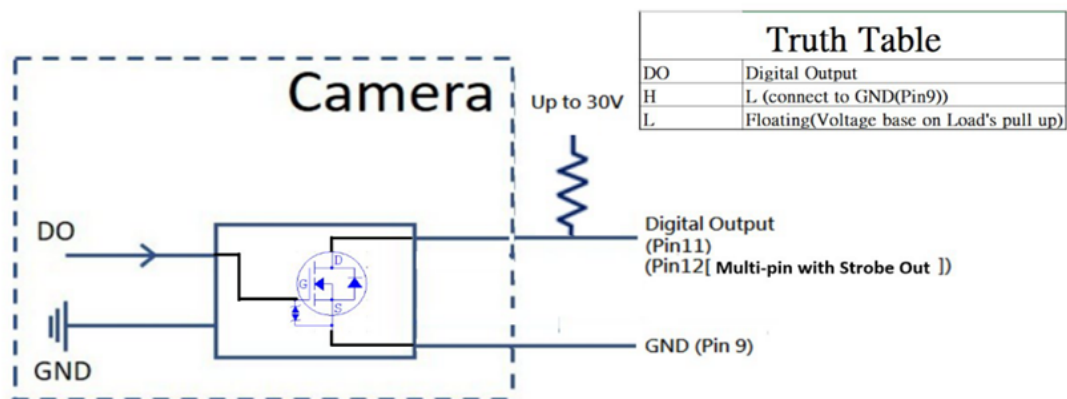
**S/W Interrupt**

Yes

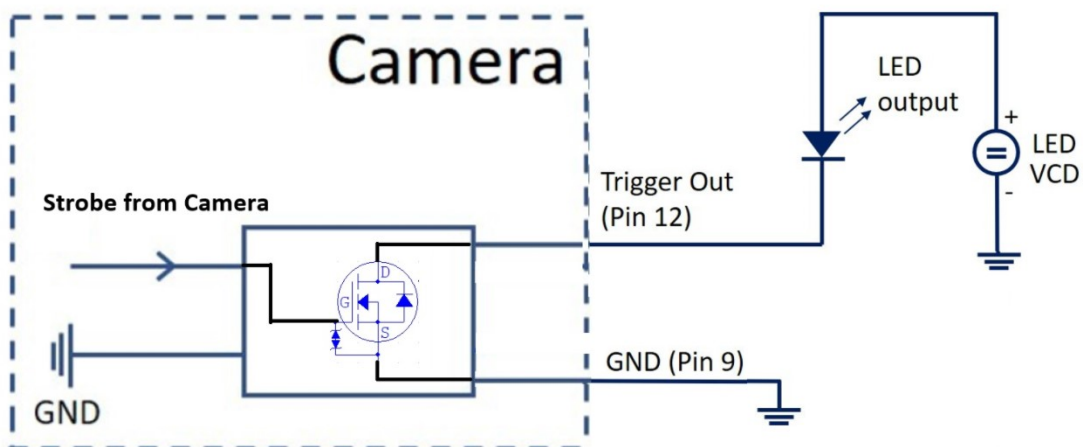


**Table 3.5: Digital Output**

| Pin No. | Sink Voltage | Sink Current  | Frequency | Response Time | S/W Interrupt |
|---------|--------------|---------------|-----------|---------------|---------------|
| 11      | Up to 30V    | Typical: 0.2A | Max. 60Hz | Max: 8us      | No            |

**Table 3.6: Strobe Out**

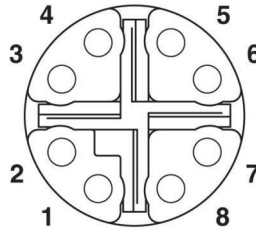
| Pin No. | Sink Voltage | Sink Current  | Frequency | Response Time | S/W Interrupt |
|---------|--------------|---------------|-----------|---------------|---------------|
| 12      | Up to 30V    | Typical: 0.2A | Max. 60Hz | Max: 4.5us    | No            |



### 3.2.3 Ethernet/LAN

#### Ethernet Connector (LAN)

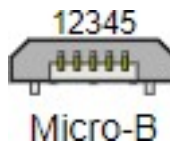
ICAM-300 series is equipped with one Ethernet controllers that are fully compliant with IEEE 802.3u 10/100/1000 Mbps CSMA/CD standards. The Ethernet port provides a M12 connector.



**Table 3.7: Ethernet Connector (LAN)**

| Pin | Signal  |
|-----|---------|
| 1   | MDI_0_P |
| 2   | MDI_0_N |
| 3   | MDI_1_P |
| 4   | MDI_1_N |
| 5   | MDI_3_P |
| 6   | MDI_3_N |
| 7   | MDI_2_N |
| 8   | MDI_2_P |

### 3.2.4 Micro USB



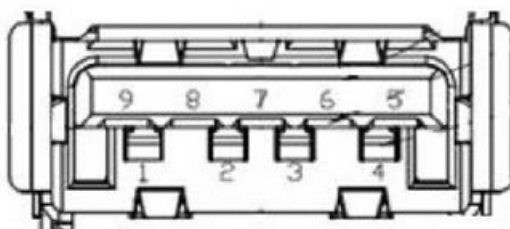
**Table 3.8: Micro USB**

| Pin | Signal           |
|-----|------------------|
| 1   | Vbus (4.4-5.25V) |
| 2   | D-               |
| 3   | D+               |
| 4   | ID               |
| 5   | GND              |

### 3.2.5 USB 3

#### USB Type-A connector

The ICAM-300 provides one USB Type-A interface connector which is only for keyboard and mouse usage. Connecting to USB storage will increase the power consumption and will heat up the system. Please refer to the table below for pin assignments.



**Table 3.9: USB Type-A Connector**

| Pin | Signal      |
|-----|-------------|
| 1   | VBUS        |
| 2   | D-          |
| 3   | D+          |
| 4   | GND         |
| 5   | StdA_SSRX-  |
| 6   | StdA_SSRX+  |
| 7   | GND_DRAIN   |
| 8   | StdA_SSRTX- |
| 9   | StdA_SSRTX+ |
| H1  | PTH_1       |
| H2  | PTH_2       |



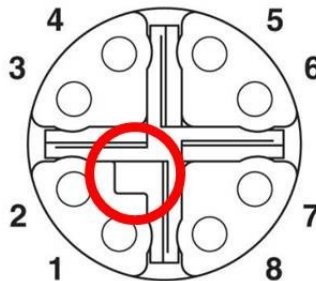
# Chapter 4

Getting Started

## 4.1 Getting Started

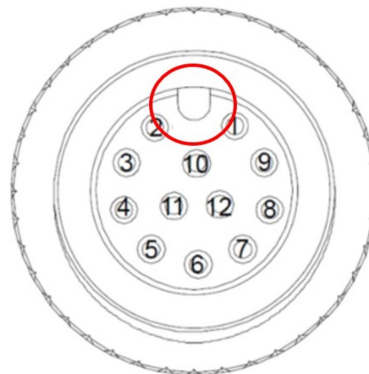
This chapter describes connection, configuration and start acquisition from ICAM-300 via host PC.

## 4.2 Connection an Ethernet Cable



1. Align the alignment pin (Cable side) with the alignment channel (Device side)
2. Inset the cable connector and tighten the threaded collar to fix the connection.

## 4.3 Connect a Power and D I/O Cable



1. Align the alignment pin (Cable side) with the alignment channel (Device side)
2. Inset the cable connector and tighten the threaded collar to fix the connection.
3. Plug in the power source after the M12 connector feast on the ICAM-300 otherwise that will damage the ICAM-300.

## 4.4 Access the ICAM-300

The default user account and the password is icam-300.

You can access the ICAM-300 to start camera preview and configuration through the Web Utility using a browser on a Windows 10 (or later) host PC.

Before configuring the ICAM-300, make sure the host PC is properly connected to the ICAM-300 in the same local network via an Ethernet cable.

The default IP address of the ICAM-300 is 192.168.0.100. Manually configure the host PC's TCP/IPv4 settings as follows:

1. IP address: 192.168.0.X (where X is any number between 1–254, avoiding IP conflicts)
2. Subnet mask: 255.255.255.0
3. Open a web browser.
4. Enter the ICAM-300 IP address in the address bar: 192.168.0.100:5001.

The ICAM-300 Web Utility interface will be displayed in the browser. For detailed parameter settings and definitions, please refer to the ICAM-300 Programming Guide.

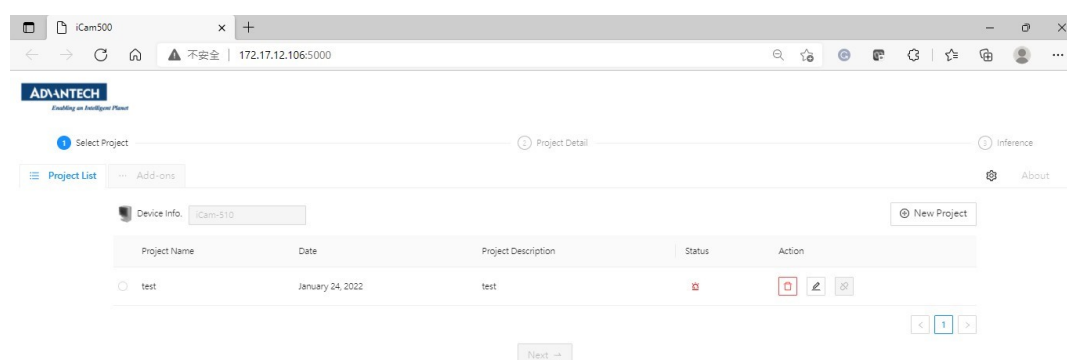


Figure 4.1 Access ICAM-300 via web browser

### 4.4.1 Image Acquisition

The camera performs continuous acquisition until an acquisition stop command is executed. The table below shows the relationship between resolutions and different color formats through the CamNavi SDK.

| Table 4.1: fps in different resolution |                   |
|----------------------------------------|-------------------|
| Resolution                             | NV12 Color Format |
| 2592x1944                              | 41                |
| 1920x1080                              | 59                |
| 1280x720                               | 59                |

## 4.4.2 Set up the Digital I/O Control

ICAM-300 provides 2 port digital input and 2 port digital output for external connection.

### 4.4.2.1 Strobe Output

DO1 (Pin 12) in the ICAM-300 is programmable as strobe out or DO.

The ICAM-300 provides 2 port digital input and 1 port digital output for external connection.

## 4.5 ICAM-300 Camera Parameters and SDK

For information on the APIs and examples please refer to the programming guide.

**Table 4.2: ICAM-300 Camera Parameters and SDK**

| Function Name         | Parameters                                                                                                                                            |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | ICAM-300-21W                                                                                                                                          |
| Trigger Mode Settings | Continuous Mode<br>S/W Trigger Mode<br>Frame rate                                                                                                     |
| Pixel Format          | NV12                                                                                                                                                  |
| Image Adjustments     | Color Sharpness (0 ~ 6)<br>Color Saturation (0 ~ 10)<br>Brightness (-12 ~ 12)<br>Contrast (1 ~ 10)<br>White balance mode (0 ~ 10)<br>ISO mode (0 ~ 8) |
| Auto Exposure Mode    | Exposure time                                                                                                                                         |
| Auto Focus Mode       | off/auto/macro/continuous                                                                                                                             |
| Platform Settings     |                                                                                                                                                       |
| I/O Settings          | DI 0/1<br>– Edge Trigger (Rising/Falling/Both)<br>DO 0/1 (*)<br>– Output High/Low<br>* Note: DO1 could switch to strobe out                           |
| User-defined LED      | Green Orange Yellow                                                                                                                                   |

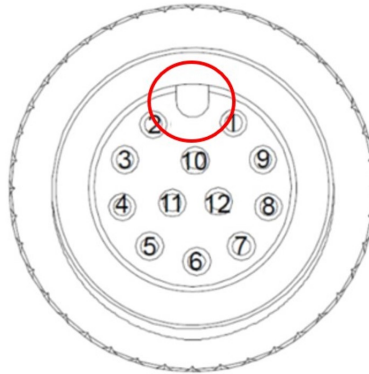
# Chapter 5

## Troubleshooting

## 5.1 Troubleshooting

Please setup the ICAM-300 with the following instructions to operate ICAM-300 in local mode for troubleshooting. If the user has problems connecting the ICAM-300 via a host PC, or if ICAM-300 still can't boot up and make an acquisition after reset up the ICAM-300, contact Advantech FAE for technical support.

## 5.2 Reconnect a Power and D I/O Cable



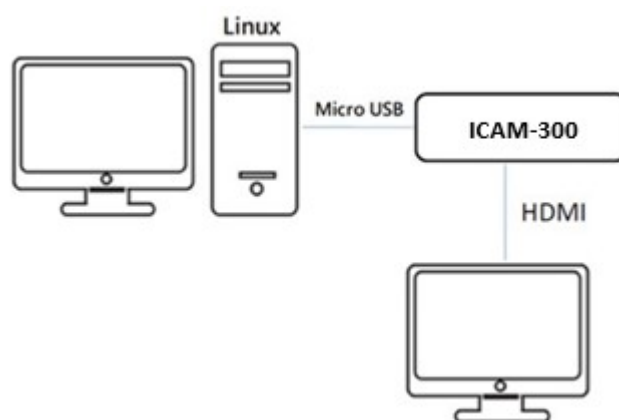
1. Align the alignment pin (Cable side) with the alignment channel (Device side).
2. Inset the cable connector and tighten the threaded collar to fix the connection.
3. Plug in the power source after the M12 connector is fastened on the ICAM-300.

## 5.3 Operate the Camera Function

1. Use Ethernet to connect with the ICAM-300.
2. Enter the default IP address of the ICAM-300: 192.168.0.100:5001.
3. The ICAM-300 web server interface will be displayed in the browser.

## 5.4 BSP Image Flashing

1. Connect ICAM-300 device and Host PC (x86 with Ubuntu 22.04 OS with 80GB or above storage) with micro USB cable. Do NOT connect USB type-A during flashing.



2. Copy BSP file (Ex: ICAM-300\_ORIN-NX\_1.0.0\_GA\_1.3.tbz) to Host PC.
3. Enter the following command on the Host PC to un-tar the BSP image file: `tar zxvf ICAM-300_qcom-multimedia-image-emmc_4.0.0.1.tar.gz`.
4. Configure the udev USB rules for the Qualcomm manufacturing vendor ID 05c6 on the Host PC. Create file `/etc/udev/rules.d/51-qcom-usb.rules` with the following content:

---

```
SUBSYSTEMS=="usb", ATTRS{idVendor}=="05c6", ATTRS{idProduct}=="9008",
MODE="0666", GROUP="plugdev"
```

---

5. Restart udev service: `sudo systemctl restart udev`.
6. With micro USB connection, press and hold the Recovery button on ICAM-300, then turn on the power.
7. Release the Recovery button after waiting for 3 seconds.
8. Enter the "lsusb" command on the Host PC.
9. ID 05c6:9008 Qualcomm...(QDL mode). means device is in recovery mode.

```
Bus 001 Device 004: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL mode)
```

10. Ensure that the ModemManager tool is not running. Stop ModemManager service if exist: `sudo systemctl stop ModemManager`.
11. Download the latest version of the QDL tool from [https://softwarecenter.qualcomm.com/#/catalog/item/Qualcomm\\_Device\\_Loader](https://softwarecenter.qualcomm.com/#/catalog/item/Qualcomm_Device_Loader) and unzip the contents of the downloaded folder.
12. On the Host PC, enter the decompressed BSP directory and type the following command:  
`sudo chmod +x qdl`  
`sudo ./qdl prog_firehose_dds.elf rawprogram*.xml patch*.xml`.

13. After 10 to 20 minutes, you should see a success message as shown below on the Host PC and ICAM-300 will automatically restart.

```
[PROGRAM] flashed "system" successfully at 44716kB/s
LOG: INFO: Calling handler for program
[PROGRAM] flashed "efi" successfully at 43788kB/s
LOG: INFO: Calling handler for program
[PROGRAM] flashed "dtb_a" successfully at 1484kB/s
...
Update Backup Header with CRC of Backup Header.
LOG: INFO: Calling handler for patch
LOG: INFO: Calling handler for setbootablestoragedrive
LOG: INFO: Using scheme of value= 1
partition 1 is now bootable
LOG: INFO: Calling handler for power
LOG: INFO: Will issue reset/power off 100 useconds, if this hangs check if watchdog is enabled
LOG: INFO: bsp_target_reset() 1
```

14. To find BSP version, access ICAM-300 from the Host PC using ssh:  
ssh icam-300@192.168.0.100  
(password: icam-300)  
Enter "cat /etc/os-release"/

```
root@adv-icam300-qcs6490:~# cat /etc/os-release
ID=qcom-wayland
NAME="Qualcomm Linux"
VERSION="1.3-ver.1.1"
VERSION_ID=1.3-ver.1.1
PRETTY_NAME="Qualcomm Linux 1.3-ver.1.1"
IMAGE_VERSION="icam300-qcs6490_v0.0.4"
KERNEL_VERSION="Linux version 6.6.52"
IMAGE_BUILD_DATE="20250728"
```



## **[www.advantech.com](http://www.advantech.com)**

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